STATS 201A Lecture 15

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Population : μ, τ, σ^2 are fixed
sample $\hat{y}, \hat{\tau}, s^2$ are random variable

$$\bar{y} \approx N(\mu, \frac{N-n}{N}\frac{\sigma^2}{n})$$

if n and $N-n$ are large

$$\sigma^2 = \frac{1}{n-1}\sum_{i=1}^{N}(y_i - \mu)^2$$

Population: $y_1, \ldots, y_n$
Probability: $\frac{1}{n}, \ldots, \frac{1}{n}$

$$Z = \frac{1}{N}\sum_{i=1}^{N} y_i$$

$$\mu = \frac{1}{N}\sum_{i=1}^{N} y_i$$